webfleet

Webfleet CO_2 Report

Accurate, user-friendly carbon emission reporting for fleets.

BRIDGESTONE
Solutions for your journey

Accurate overview of your fleet's direct CO_2 emissions

Using a methodology that's certified by TÜV Rheinland[1], it provides detailed information on how much CO_2 your vehicles produce. As well as helping you to meet the demands of regulations such as the EU's new Corporate Sustainability Reporting Directive (CSRD), it gives you visibility over how, where and why your fleet is producing unnecessary CO_2.

Contents

1. Introduction
   - Climate change
   - Impact of transport
   - Transport regulation
   - EU Corporate Sustainability Reporting requirements
   - Reporting requirements
   - Fleets and reporting: Scopes are determined by vehicle ownership
   - Risk of non-compliance
   - Calculating emissions from transport: GHG Protocol Standard
2. Summary
   - Value propositions
   - What is CO_2 and why is visualization important?
3. Features
   - Webfleet reports
   - Webfleet CO_2 Report
   - Webfleet CO_2 Report for compliance
   - Example of Webfleet CO_2 Report
   - Webfleet CO_2 Report: Two calculation methods
   - Webfleet CO_2 Report: Fuel-based calculation method
   - Webfleet CO_2 Report: Distance-based method
   - Webfleet CO_2 Report: Manual avgerage fuel consumption adjustment
   - Webfleet CO_2 Report: Vehicle type selection
   - Webfleet CO_2 Report: Fuel type selection
   - Trip mode separation and requirements
   - EV CO_2 Reporting frame
   - National grid values
   - EV CO_2 Reporting calculating formula
   - EV CO_2 Reports
   - Average energy consumption
   - Energy-based calculation example
   - Distance Based calculation for EV
4. Benefits
   - Customer benefits
5. Requirements and price
   - Tariff requirements
6. Appendix
   - FAQs

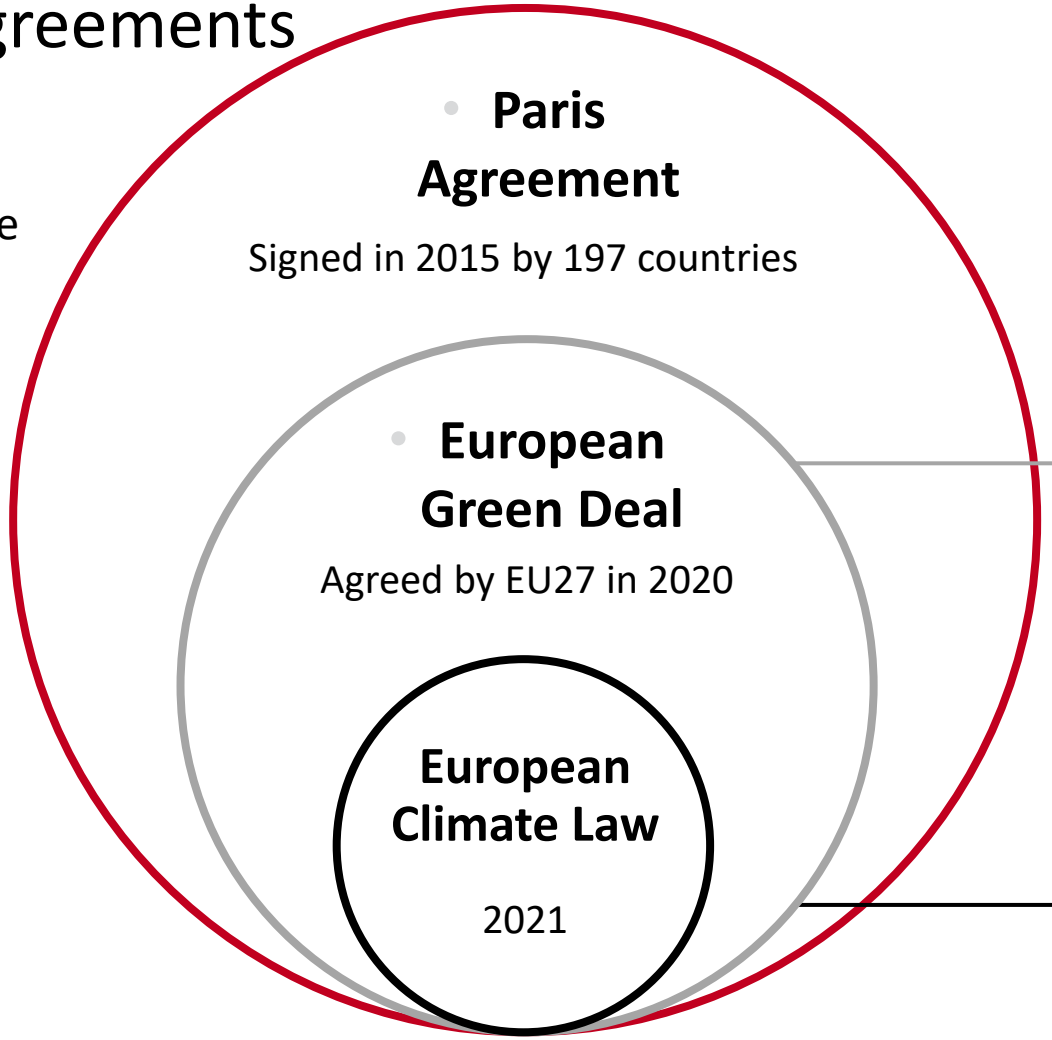
Contents

1. Introduction
   - Climate change
   - Impact of transport
   - Transport regulation
   - EU Corporate Sustainability Reporting Directive requirements
   - Reporting requirements
   - Fleets and reporting
   - Risk of non-compliance
   - Calculating emissions from transport: GHG Protocol Standard
2. Summary
   - Value propositions
   - What is CO_2 and why is visualization important?
3. Features
   - Webfleet reports
   - Webfleet CO_2 Reports
   - Webfleet CO_2 Report for compliance
   - Example of Webfleet CO_2 Report
   - Webfleet CO_2 Report: Two calculation methods
   - Webfleet CO_2 Report: Fuel-based calculation method
   - Webfleet CO_2 Report: Distance-based calculation method
   - Webfleet CO_2 Report: Manual average fuel consumption adjustment
   - Webfleet CO_2 Report: Vehicle type selection
   - Webfleet CO_2 Report: Fuel type selection
   - Trip mode separation and requirements
   - Webfleet CO_2 emissions for EVs
   - National grid values
   - CO_2 emission calculating formula for EV
   - Average energy consumption
   - Energy-based calculation example
   - Distance Based calculation for EV
4. Benefits
   - Customer benefits
5. Requirements and price
   - Tariff requirements
6. Appendix
   - FAQs

Climate change

Overarching climate agreements

- Reduce global GHG emissions to limit global temperature increase to 2°C
- Pursue 1.5°C limit
- **Reduce CO_2 emissions by 45%** by 2030 compared to 2010
- Net zero emissions by 2050



- Reduce emissions by 55% in 2030 compared to 1990
- Climate neutral continent by 2050
- Set Green Deal objectives into law

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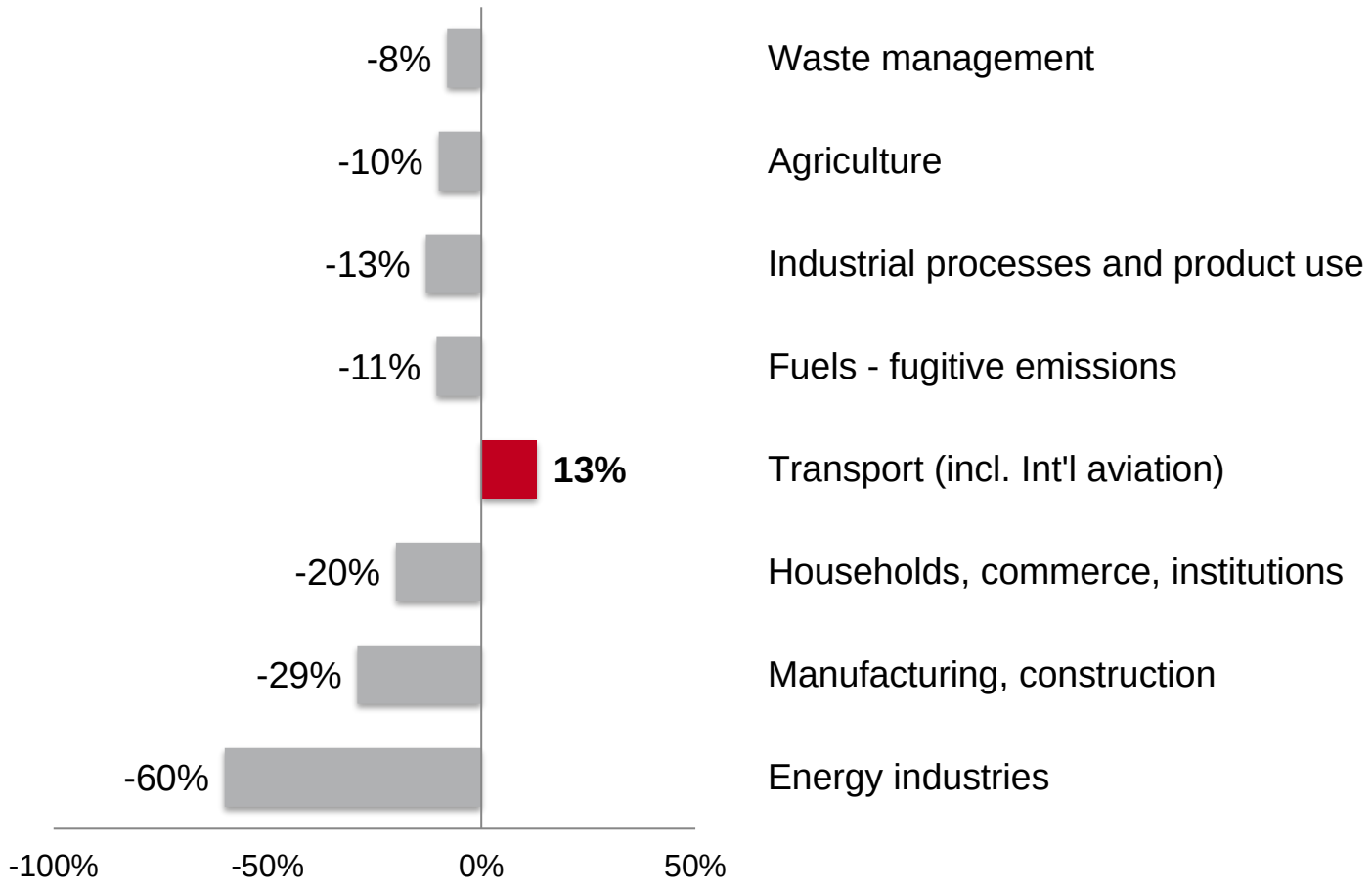
Impact of transport

GHG emissions by sector 1990-2021



Transport is the only sector in the EU showing an increase in GHG emissions

Source: Eurostat, Greenhouse gas emissions by source sector (source: EEA), online data code env_air_gge__custom_8301478

Transport regulation

EU standards for new vehicles

	2020	2025	2030	2035	2040
Cars	95 g CO_2/km	15% reduction	55% reduction	100% reduction	
Vans	147 g CO_2/km	15% reduction	50% reduction	100% reduction	
Heavy-duty vehicles: New targets			45% reduction	65% reduction	90% reduction
Heavy-duty vehicles: Previous		15% reduction	30% reduction		

Source: European Commission CO_2 emission performance standards for cars and vans

EU Corporate Sustainability Reporting requirements

The **Non-Financial Reporting Directive (NFRD**) requires **listed companies with more than 500 employees** to include **non-financial statements** as an integral part of their **annual public reporting** obligations. The NFRD is to be replaced by CSRD from January 2023.

The **Corporate Sustainability Reporting Directive (CSRD)** is a new EU regulation that came into play in January 2024. It requires certain companies to report on, amongst other things, the impact of their activities on people and the environment. Even if your business is not required to report its CO_2 emissions, you may still be impacted by the directive's demand for more transparency around carbon emissions. The **European Sustainability Reporting Standard (ESRS)** support this with detailed reporting requirements.

The **EU Taxonomy** is a classification system that defines activities deemed to be environmentally sustainable. As part of the CSRD, companies are required to report under the EU Taxonomy regulation in addition to the ESRS. Including companies and financial institutions.

While different types of businesses will be affected by the regulation at different times, over the coming years many **thousands of companies across the continent will need to report on their CO_2 emissions for the entire fiscal year**.

Reporting requirements

Who is affected and when?

Non-Financial Reporting Directive (NFRD)

Corporate Sustainability Reporting Directive (CSRD)

EU Taxonomy

2020 **2021** **2022** **2023** **2024** **2025** **2026** **2027**

- Large companies
- Listed companies with more than 500 employees

Listed companies with > 500 employees to start recording CO_2 emissions from Jan 2024 and to start reporting in 2025

Other large companies meeting two out of three criteria

> 250 employees
> €50M revenue
> €25M total assets

Listed SMEs meeting two out of three criteria

> 10 employees
> €900k revenue
> €450k total assets

Companies with a non-EU parent

> €150M EU turnover

- **Companies in scope for mandatory reporting** 11,000 under NFRD
- **50,000 under CSRD**

Source: European Commission, Corporate sustainability reporting, last seen online on 5 November 2023.

Fleets and reporting: Scopes are determined by vehicle ownership

Webfleet CO_2 Report

Scope 1

Direct GHG emissions from **owned** or controlled fuel engine **vehicles**

Scope 2

Indirect GHG emissions from purchased **electricity** used by **owned** or controlled **electric vehicles**

Scope 3

GHG emission from upstream and downstream transportation on **non-owned**, not controlled **vehicles**, including:

- Inbound- and outbound logistics
- Employee Commuting
- Business Travel

Company owned vehicles

Non-owned vehicles

e.g. leased vehicles, transport services, outsourced and sub-contracted activities

Risks of non-compliance

Consequences of not complying with CSRD

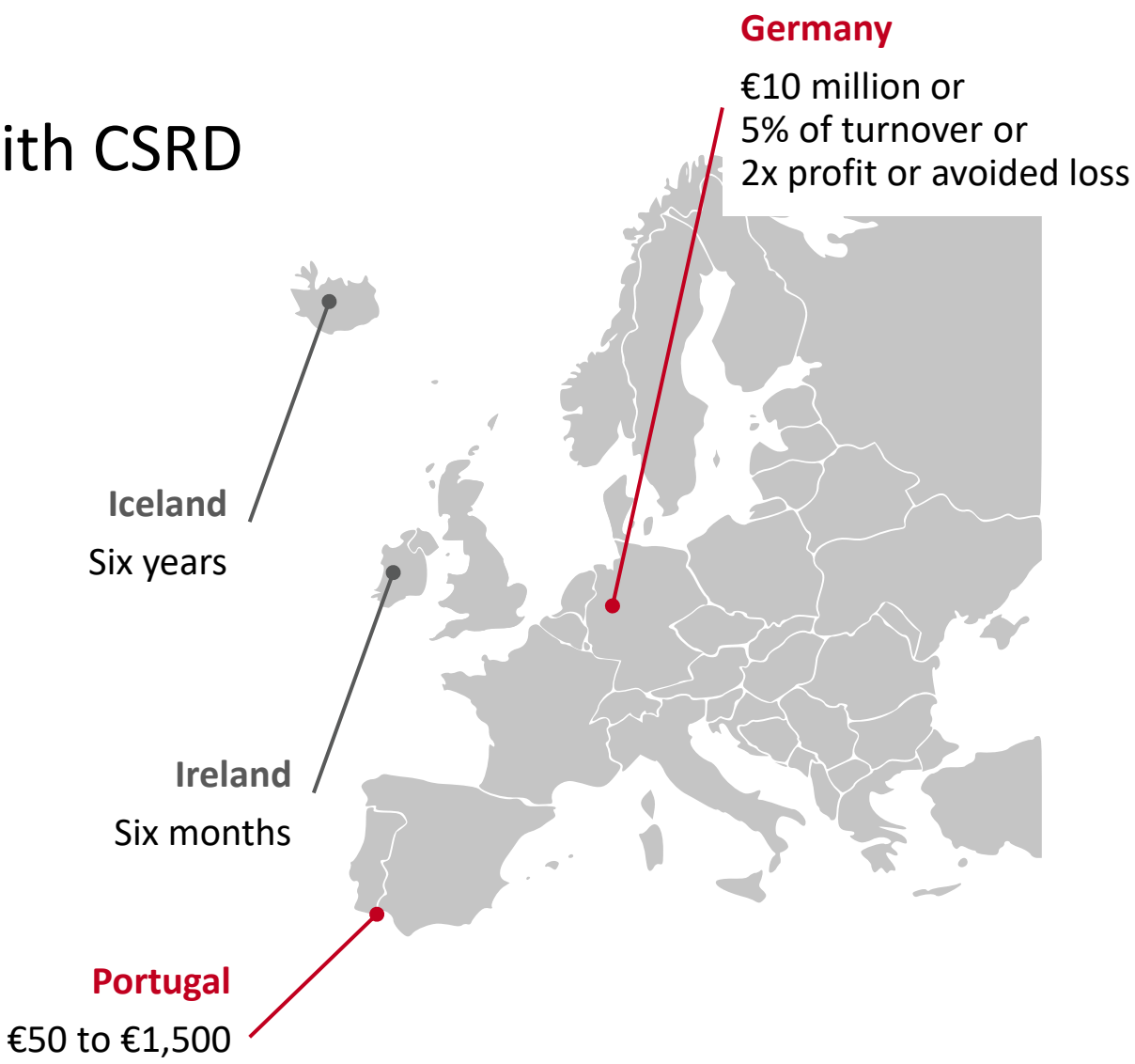
All EU27 countries state penalties for non-compliance

Fines differ by country

Prison sentences exist too

Other risks

- Reputational damage
- Loss of stakeholder confidence
- Legal action from non-government entities



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Calculating emissions from transport: GHG Protocol Standard

Webfleet CO_2 Report

Fuel-based method

- Measures actual **average fuel consumption** from vehicle electronics.
- Applies appropriate emission factors to calculate level of CO_2 emissions.
- The **best estimate of CO_2 emissions**.

Distance-based method

- Measures **distance, type of vehicle and average fuel consumption.**
- Applies the appropriate emission factors (such as fuel type) for the vehicle to calculate level of CO_2 emissions.

Spend-based method

- Calculates the amount of **money spent** on **each mode of business travel** transport and secondary emission factors.

Value Proposition

	Who	Challenges	Benefits	Proof points and value messages
Fleet type: All vehicles **Key verticals:** • All vertical **Decision makers:** • Owner • Sustainability Manager • ESG office • Operations Manager • Financial Director • Fleet Manager	• Owner • Fleet manager	• I need to record the CO_2 of every vehicle in my fleet. • I want to reduce the CO_2 of my fleet. • I want to reward good driving behaviour but have no visibility into performance.	**Track the CO_2 produced by every vehicle in your fleet** And take action to reduce your emissions	**Real-time and historical insights** • The Webfleet CO_2 Report enables you to accurately measure the CO_2 produced by your vehicles. You can see not only how much CO_2 your vehicles are emitting, but also why and where it is happening. With these real-time and historical insights, you can zero in on where the real opportunities are to cut your fleet's emissions. **The impact of driving behaviour** • Driving behaviour like idling and harsh braking can directly lead to needless fuel consumption and CO_2 emissions. With data on CO_2 emissions in your Webfleet OptiDrive 360[2] reporting, you can see specifically where drivers are needlessly increasing your carbon footprint and coach them to adopt more sustainable and cost-effective driving styles.
	• Sustainability Manager • ESG office • Financial Director	• I need to report the CO_2 of every vehicle in my fleet. • I need to lower our environmental impact. • I need to reduce my CO_2 emissions. • I need to comply with CSRD regulations. • I need to provide my CO_2 emissions to my customers or to the authorities.	**Report accurately and easily** On your business' CO_2 emissions and environmental impact **Trust a TÜV Rheinland certified calculation method[1]** For reliable and actionable insights	**Accurate report data** • With the Webfleet CO_2 Report you can rest assured you'll have all the data you need to report on your CO_2 emissions – both to the authorities and to your customers that request it. It's the only CO_2 reporting tool from a telematics provider with a methodology certified by TÜV Rheinland. So, whether you need this data to stay compliant, stay competitive or both, Webfleet has you covered.

[1] To meet state-of-the-art scientific standards, all platform-embedded calculations and decarbonisation solutions are fully aligned with internationally recognised scientific methodologies and standards such as the Greenhouse Gas Protocol and the Science Based Targets initiative (SBTi). The scientific accurateness of the applied Corporate Carbon Footprint (CCF) calculation methodology, Scope 1, is certified by TÜV Rheinland.
[2] Webfleet OptiDrive 360 is available on ECO tariff and above

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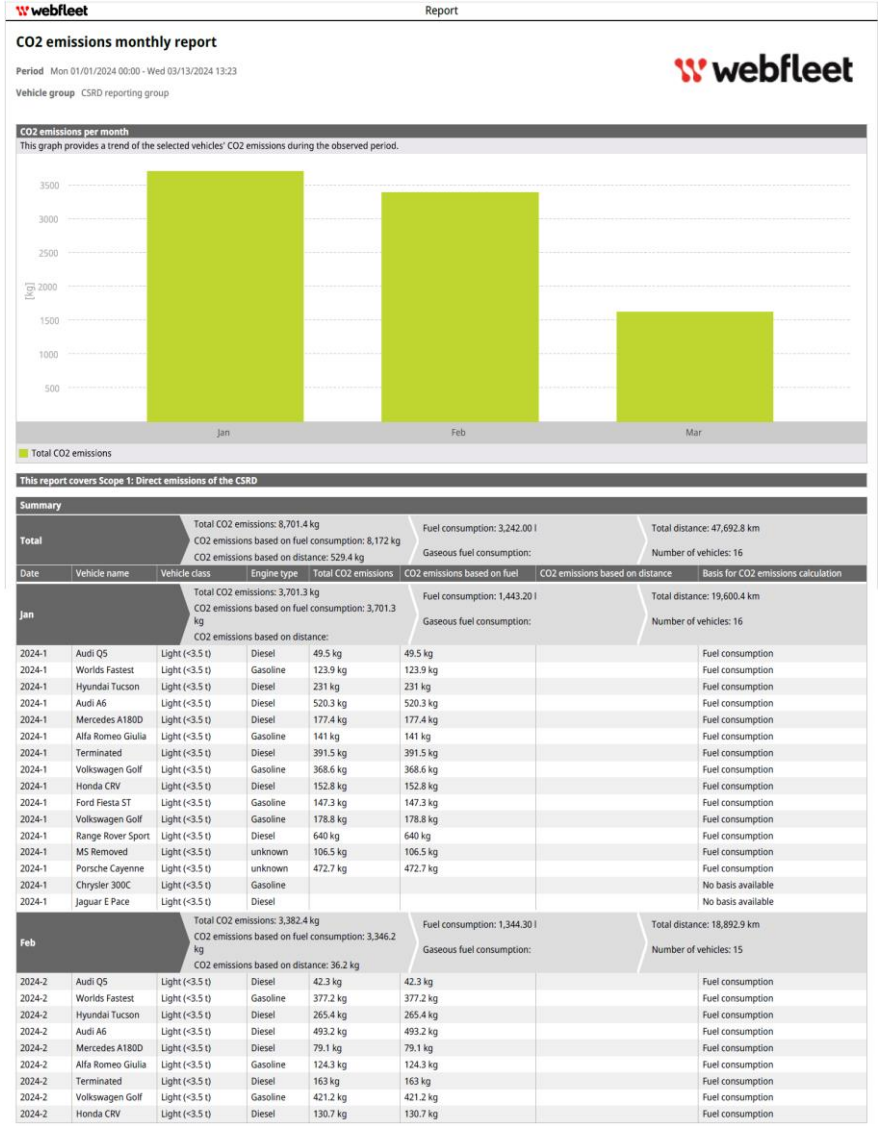
What is CO_2 and why is visualization important?

- Vehicle CO_2 typically refers to the amount of Carbon Dioxide (CO_2) emissions produced by vehicles. When fossil fuels such as gasoline or diesel are burned in an engine, carbon dioxide is one of the byproducts released into the atmosphere.
- The level of CO_2 emissions from a vehicle depends on various factors including the type of fuel used, the engine's efficiency, and the vehicle's size and weight.
- Emission reduction is a universal business driver and several of our customers have ambitious pollutant reduction targets. We can support our customer to reach their targets by helping them:
    - Encourage a more sustainable driving style e.g. reduce speed.
    - Smoothly transition from combustion engine vehicles to electric vehicles.
    - Find the most efficient and optimal combination of vehicles.
- Reporting itself does not make anyone sustainable. Reports and visualization help our customers understand the current situation and to set and reach their targets.
- Based on CSRD regulation, several of our customers must create reports starting from year 2024 pollutants. Typically, companies are publishing this kind of data in their annual report, sustainability report and/or their webpage.



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CO2 emissions monthly report

Period Mon 01/01/2024 00:00 - Wed 03/13/2024 13:23

Vehicle group CSRD reporting group

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CO2 emissions per month

This graph provides a trend of the selected vehicles' CO2 emissions during the observed period.

This report covers Scope 1: Direct emissions of the CSRD

Summary

Total	Total CO2 emissions: 8,701.4 kg CO2 emissions based on fuel consumption: 8,172 kg CO2 emissions based on distance: 529.4 kg	Fuel consumption: 3,242.00 l Gaseous fuel consumption:	Total distance: 47,692.8 km Number of vehicles: 16

Date	Vehicle name	Vehicle class	Engine type	Total CO2 emissions	CO2 emissions based on fuel	CO2 emissions based on distance	Basis for CO2 emissions calculation
Jan	Total CO2 emissions: 3,701.3 kg CO2 emissions based on fuel consumption: 3,701.3 kg CO2 emissions based on distance:				Fuel consumption: 1,443.20 l Gaseous fuel consumption:	Total distance: 19,600.4 km Number of vehicles: 16	
2024-1	Audi Q5	Light (<3.5 t)	Diesel	49.5 kg	49.5 kg		Fuel consumption
2024-1	Worlds Fastest	Light (<3.5 t)	Gasoline	123.9 kg	123.9 kg		Fuel consumption
2024-1	Hyundai Tucson	Light (<3.5 t)	Diesel	231 kg	231 kg		Fuel consumption
2024-1	Audi A6	Light (<3.5 t)	Diesel	520.3 kg	520.3 kg		Fuel consumption
2024-1	Mercedes A180D	Light (<3.5 t)	Diesel	177.4 kg	177.4 kg		Fuel consumption
2024-1	Alfa Romeo Giulia	Light (<3.5 t)	Gasoline	141 kg	141 kg		Fuel consumption
2024-1	Terminated	Light (<3.5 t)	Diesel	391.5 kg	391.5 kg		Fuel consumption
2024-1	Volkswagen Golf	Light (<3.5 t)	Gasoline	368.6 kg	368.6 kg		Fuel consumption
2024-1	Honda CRV	Light (<3.5 t)	Diesel	152.8 kg	152.8 kg		Fuel consumption
2024-1	Ford Fiesta ST	Light (<3.5 t)	Gasoline	147.3 kg	147.3 kg		Fuel consumption
2024-1	Volkswagen Golf	Light (<3.5 t)	Gasoline	178.8 kg	178.8 kg		Fuel consumption
2024-1	Range Rover Sport	Light (<3.5 t)	Diesel	640 kg	640 kg		Fuel consumption
2024-1	MS Removed	Light (<3.5 t)	unknown	106.5 kg	106.5 kg		Fuel consumption
2024-1	Porsche Cayenne	Light (<3.5 t)	unknown	472.7 kg	472.7 kg		Fuel consumption
2024-1	Chrysler 300C	Light (<3.5 t)	Gasoline				No basis available
2024-1	Jaguar E Pace	Light (<3.5 t)	Diesel				No basis available
Feb	Total CO2 emissions: 3,382.4 kg CO2 emissions based on fuel consumption: 3,346.2 kg CO2 emissions based on distance: 36.2 kg				Fuel consumption: 1,344.30 l Gaseous fuel consumption:	Total distance: 18,892.9 km Number of vehicles: 15	
2024-2	Audi Q5	Light (<3.5 t)	Diesel	42.3 kg	42.3 kg		Fuel consumption
2024-2	Worlds Fastest	Light (<3.5 t)	Gasoline	377.2 kg	377.2 kg		Fuel consumption
2024-2	Hyundai Tucson	Light (<3.5 t)	Diesel	265.4 kg	265.4 kg		Fuel consumption
2024-2	Audi A6	Light (<3.5 t)	Diesel	493.2 kg	493.2 kg		Fuel consumption
2024-2	Mercedes A180D	Light (<3.5 t)	Diesel	79.1 kg	79.1 kg		Fuel consumption
2024-2	Alfa Romeo Giulia	Light (<3.5 t)	Gasoline	124.3 kg	124.3 kg		Fuel consumption
2024-2	Terminated	Light (<3.5 t)	Diesel	163 kg	163 kg		Fuel consumption
2024-2	Volkswagen Golf	Light (<3.5 t)	Gasoline	421.2 kg	421.2 kg		Fuel consumption
2024-2	Honda CRV	Light (<3.5 t)	Diesel	130.7 kg	130.7 kg		Fuel consumption

Features

Webfleet reports

Stay on top of everything from fuel efficiency to legal compliance, so you can focus on delivering top-quality service.

Webfleet CO_2 Report

The Webfleet CO_2 Report gives you an accurate overview of your fleet's direct CO_2 emissions. It is available in Webfleet reporting at Monthly and/or Annual levels in both PDF and Excel formats. It uses the fuel and distance-based calculation methods.

Trip mode for Co2 emissions

Similarly, like in Webfleet, it is possible to separate different trip modes for CO_2 reports: business, commute and private.

Fuel-based calculation method

Uses a vehicle's fuel consumption to calculate CO_2 emissions.

Distance-based calculation method

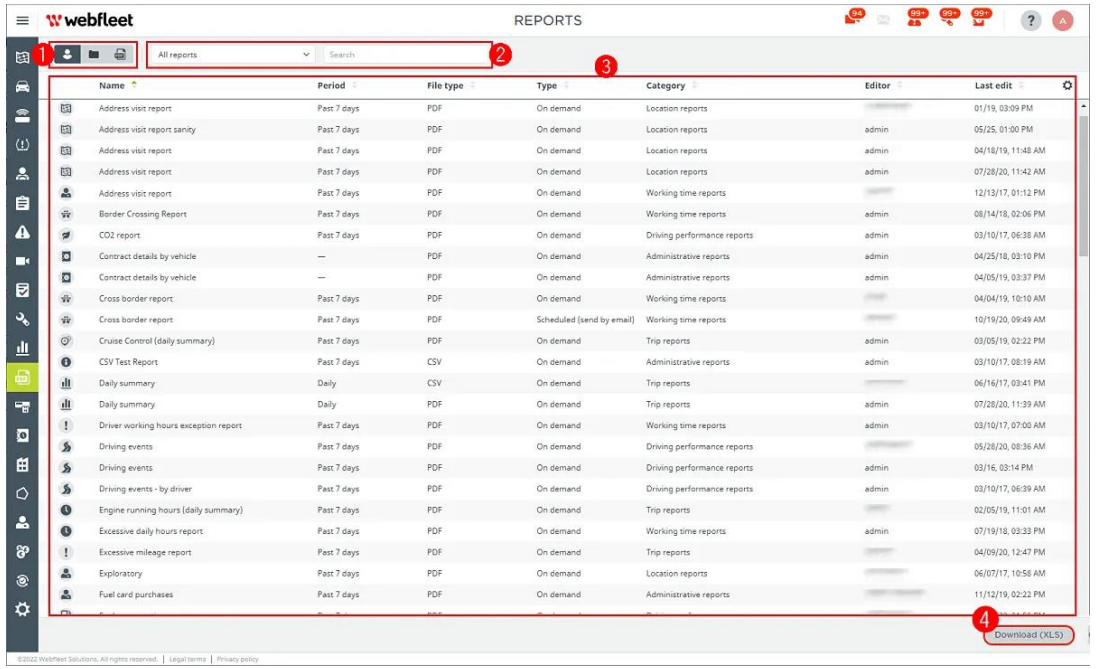
If appropriate fuel data is not available, then CO_2 calculation can be made based on distance driven.

CO_2 emissions per vehicle

Summary reports are developed mainly for Scope 1 reporting purposes. **Scope 1** is the direct emissions from **owned** or controlled **fuel-engine vehicles, while Scope 2 covers** the indirect emissions **from purchased energy**/electricity grid for **electric vehicles.**

Webfleet reports

- With Webfleet reports, you get quick overviews that enable you to identify potential improvements and stay up-to-date on all elements of fleet performance.
- You can also analyse the performance of your fleet in depth, with detailed historical information.
- You can download reports on demand or receive scheduled reports by email.

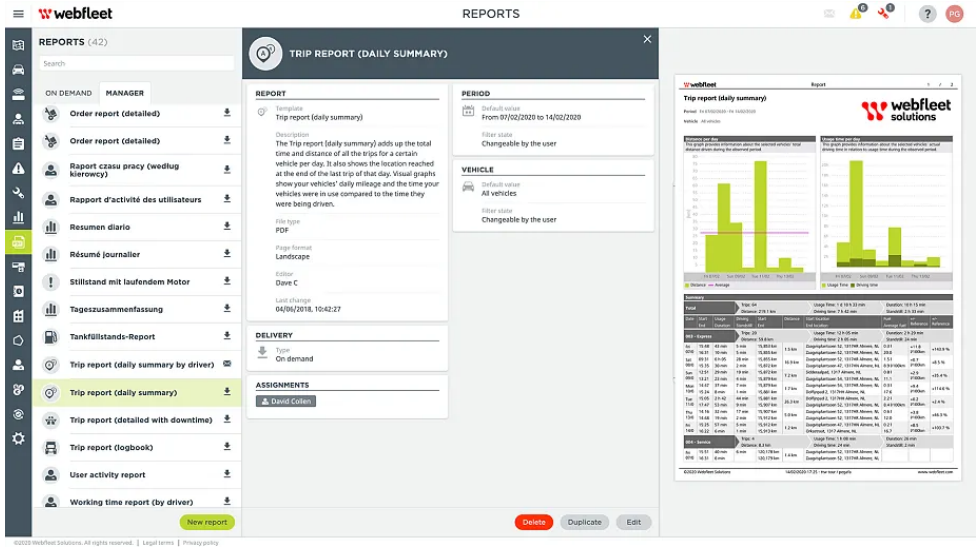


1. **View selector** – Using the view selector for the Reports view, you can switch between the following views:
   - **My reports** – Here you can download reports on demand.
   - **My archive** – Here you can download scheduled reports stored in Webfleet for later downloading.
   - **All reports** – Here you can create reports to be available for downloading on demand, from the archive or to be sent to an email address.
2. **Filters** – You can filter the list of reports by report category and free-text.
3. **List of reports** – The list of reports shows all reports that are available. Select a report from the list for downloading.
4. **Download (XLS)** - Click this button to download an XLS file showing the currently filtered table.

Note: Depending on the Webfleet subscription, individual configuration, and user profile, you may not have all features and functionality described.

Webfleet reports

- With Webfleet reports, you stay on top of everything from fuel efficiency to legal compliance. That enables you to focus on delivering top quality service.
- Comprehensive Overview reporting gives you the top-level view, without having to click through single reports individually. To help you get to the information you need fast, we suggest a selection of the most popular fleet management reports used by our customers.
- The most popular fleet management reports:
  - Trip report (daily summary)
  - Trip report (logbook)
  - OptiDrive 360 report
  - **The Webfleet CO_2 Report**
  - Fuel consumption report
  - Maintenance report
  - Working time report (by driver)
  - And many more



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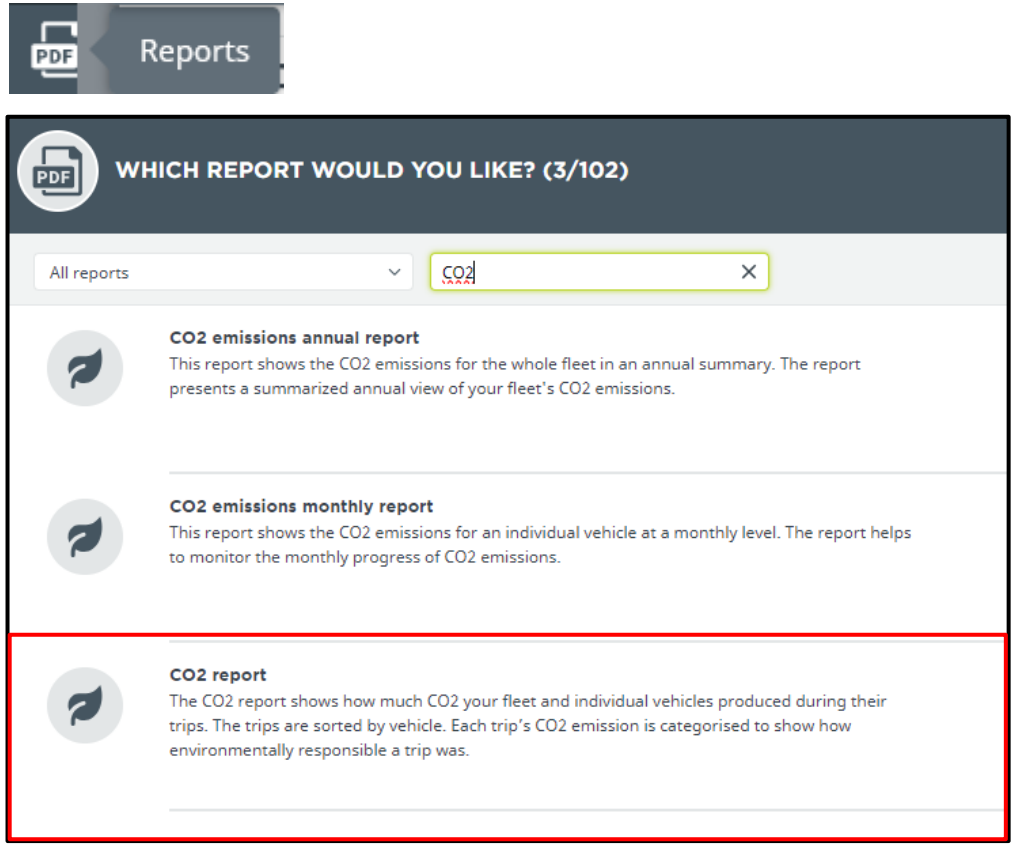
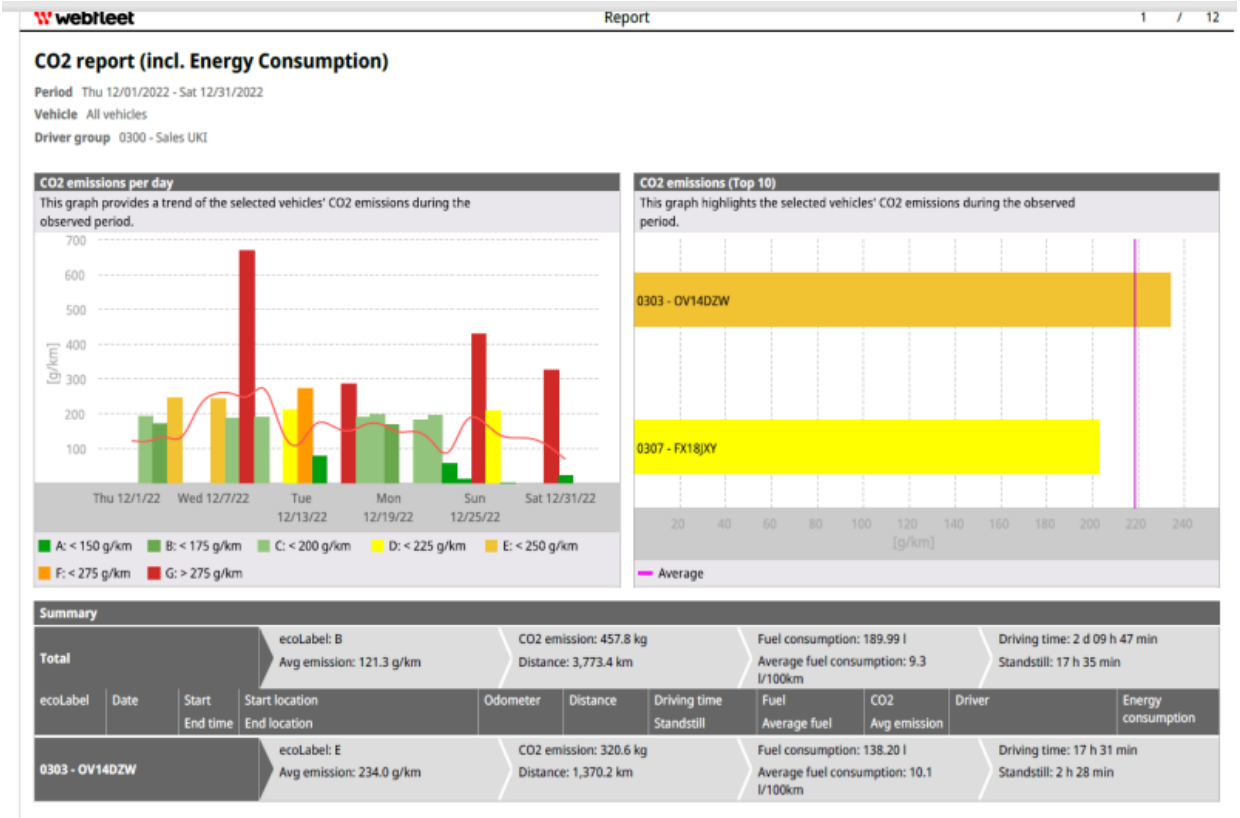
Webfleet CO_2 Report

- The Webfleet CO_2 Report gives you an accurate overview of your fleet's carbon emissions.
- There are **three different CO_2 reports** in Webfleet:
  - **The CO_2 report:** It shows how much CO_2 your fleet and individual vehicles produced during trips.
  - **The CO_2 emissions monthly report**: It shows the CO_2 emissions for an individual vehicle at a monthly level.
  - **CO_2 emission annual report**: It shows the CO_2 emissions for the whole fleet in an annual summary.



Webfleet CO_2 Report

- **The Webfleet CO_2 Report** is the standard trip level CO_2 report.
- You can easily see how much CO_2 your fleet and individual vehicles produced during their trips.
- The trips are sorted by vehicle. Each trip's CO_2 emissions are categorized to show how environmentally responsible a trip was.
- This **CO_2 report** offers reports in trip level, which is good for individual deep dives on a driver or a trip.
- The time horizon is limited to one month.
- This report has been in the Webfleet platform for many years already.



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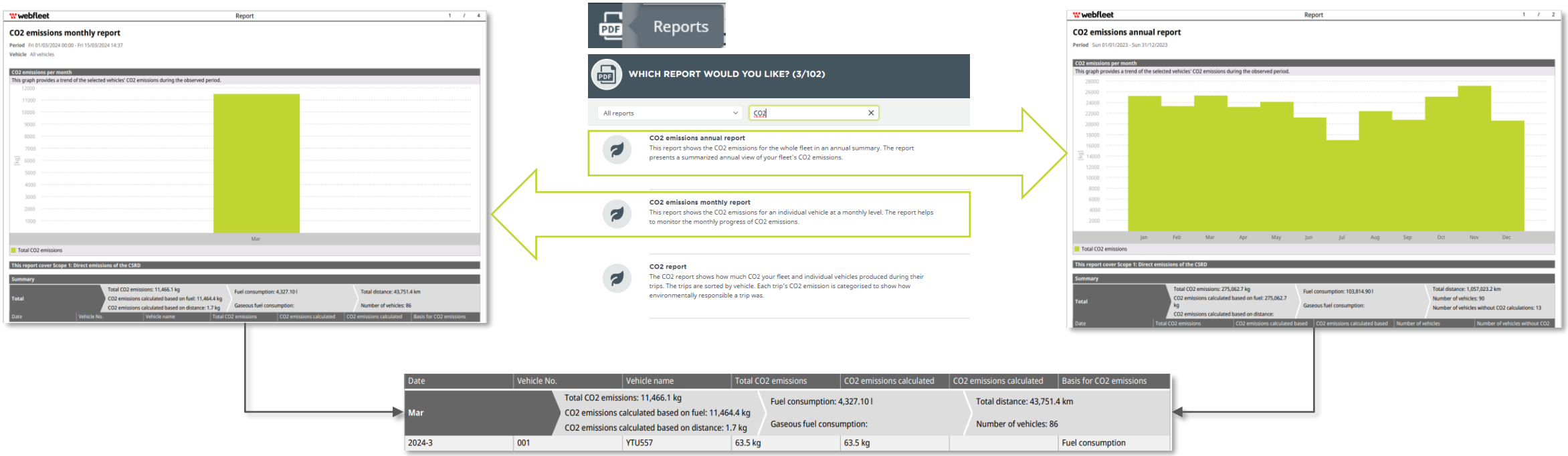
Webfleet CO_2 Report for compliance

- This report helps you to manage your reporting on direct emissions from owned or controlled fuel-engine vehicles (Scope 1). It provides
  - **Monthly summary:** The CO_2 emission produced by every vehicle in your fleet.
  - **Annual summary:** A full calendar year overview of your whole fleet.
- It is available in both PDF and Excel format for further data analyses and/or reporting purposes.



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Example of Webfleet Co2 Report Scope 1 and 2

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Summary

Total

Total CO2 emissions: 12,351.2 kg
CO2 emissions based on fuel consumption: 12,077.7 kg
CO2 emissions based on distance: 273.5 kg
Scope 1 CO2 emissions: 12,188.1 kg

Total fuel consumption: 4,540.00 l
Gaseous fuel consumption:
Energy consumption: 321.7 kWh
Scope 2 CO2 emissions: 163.1 kg

Total distance: 98,285.7 km
Number of vehicles: 19

Jun

Total CO2 emissions: 12,351.2 kg
CO2 emissions based on fuel consumption: 12,077.7 kg
CO2 emissions based on distance: 273.5 kg
Scope 1 CO2 emissions: 12,188.1 kg

Total fuel consumption: 4,540.00 l
Gaseous fuel consumption:
Energy consumption: 321.7 kWh
Scope 2 CO2 emissions: 163.1 kg

Total distance: 98,285.7 km
Number of vehicles: 19

Date	Vehicle no.	Vehicle name	Total CO2 emissions	Scope 1 CO2 emissions	Scope 2 CO2 emissions
2024-6	[illegible]	[illegible]	68.6 kg	68.6 kg	
2024-6	[illegible]	[illegible]	71 kg	71 kg	
2024-6	[illegible]	[illegible]	177.1 kg	177.1 kg	
2024-6	[illegible]	[illegible]	37.5 kg		37.5 kg
2024-6	[illegible]	[illegible]	253.8 kg	253.8 kg	
2024-6	[illegible]	[illegible]	4.1 kg	4.1 kg	
2024-6	[illegible]	[illegible]	1,727 kg	1,727 kg	
2024-6	[illegible]	[illegible]	1,746.2 kg	1,746.2 kg	
2024-6	[illegible]	[illegible]	1,807.7 kg	1,807.7 kg	
2024-6	[illegible]	[illegible]	1,952.4 kg	1,952.4 kg	
2024-6	[illegible]	[illegible]	2,043.6 kg	2,043.6 kg	
2024-6	[illegible]	[illegible]	2,336.6 kg	2,336.6 kg	
2024-6	[illegible]	[illegible]	3.3 kg		3.3 kg
2024-6	[illegible]	[illegible]	17.9 kg		17.9 kg
2024-6	[illegible]	[illegible]	44.1 kg		44.1 kg
2024-6	[illegible]	[illegible]	17.9 kg		17.9 kg
2024-6	[illegible]	[illegible]	24.5 kg		24.5 kg
2024-6	[illegible]	[illegible]	17.9 kg		17.9 kg
2024-6	[illegible]	[illegible]			

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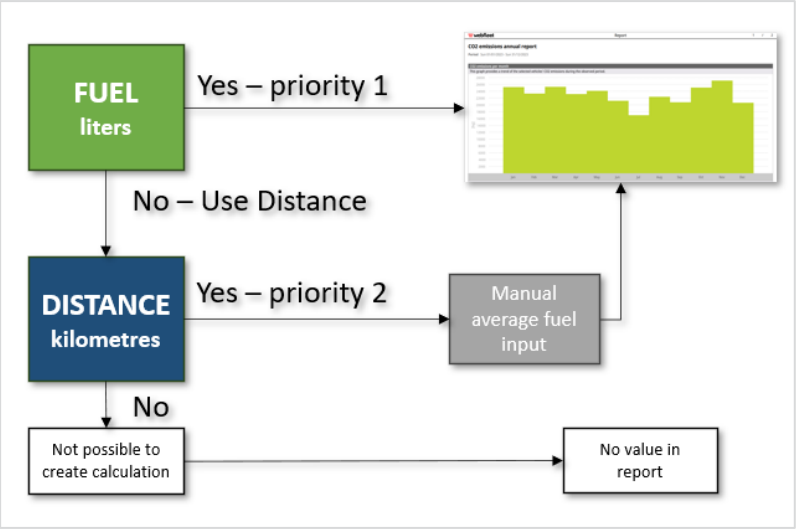
Webfleet CO_2 Report: Two calculation methods

The Webfleet CO_2 Report can use either the **fuel-based** or **distance-based** calculation methods, depending on data availability.

The basis for CO_2 emissions calculation

- ***Fuel-based method:*** If a vehicle reports fuel consumption, the CO_2 emissions are calculated based on its fuel data.
- ***Distance-based method:*** If a vehicle reports only distance, the CO_2 emissions are calculated based on its average fuel consumption.
- ***Mixed:*** Both methods can also be used together to calculate CO_2 emissions. For example, this could be the case during a month in which the tariff was upgraded when a vehicle that previously reported only distance has started providing fuel consumption data.
- ***No basis available:*** It may be the case that we cannot calculate CO_2 emissions using distance. There are two possible cases:
  - A vehicle operates using fuel other than DIESEL, PETROL and a user did not provide average fuel consumption on vehicle level (for example: CNG/LNG powered vehicles)
  - There is no distance reported at all. This could be caused by several possible reasons but is most often down to a hardware or installation issue.



Fuel consumption
Mixed
Fuel consumption
Fuel consumption
No basis available
Fuel consumption

Example from report

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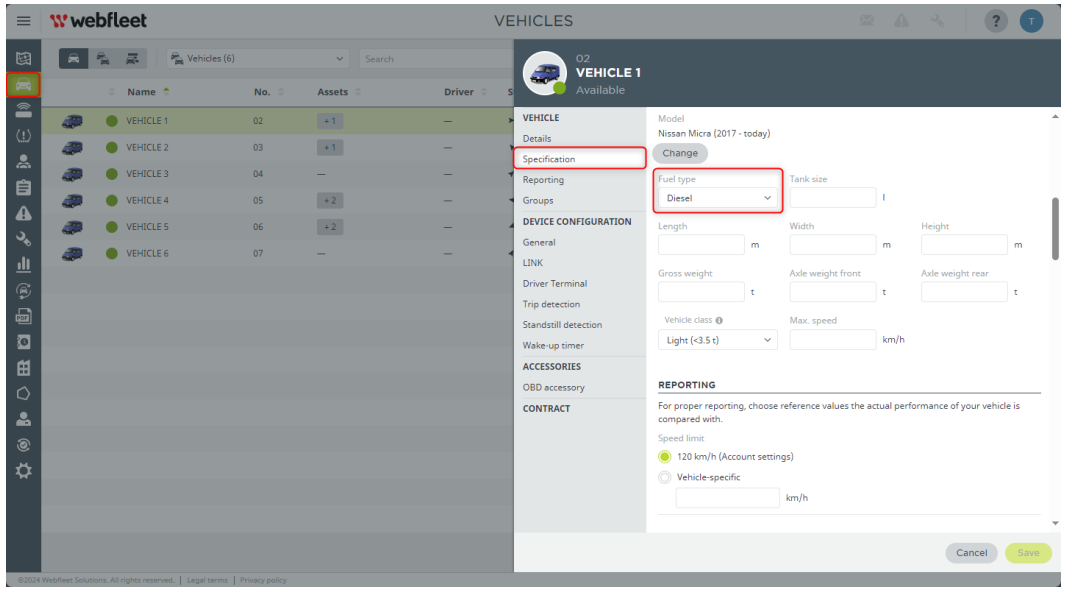
Webfleet CO_2 Report: Fuel-based calculation method

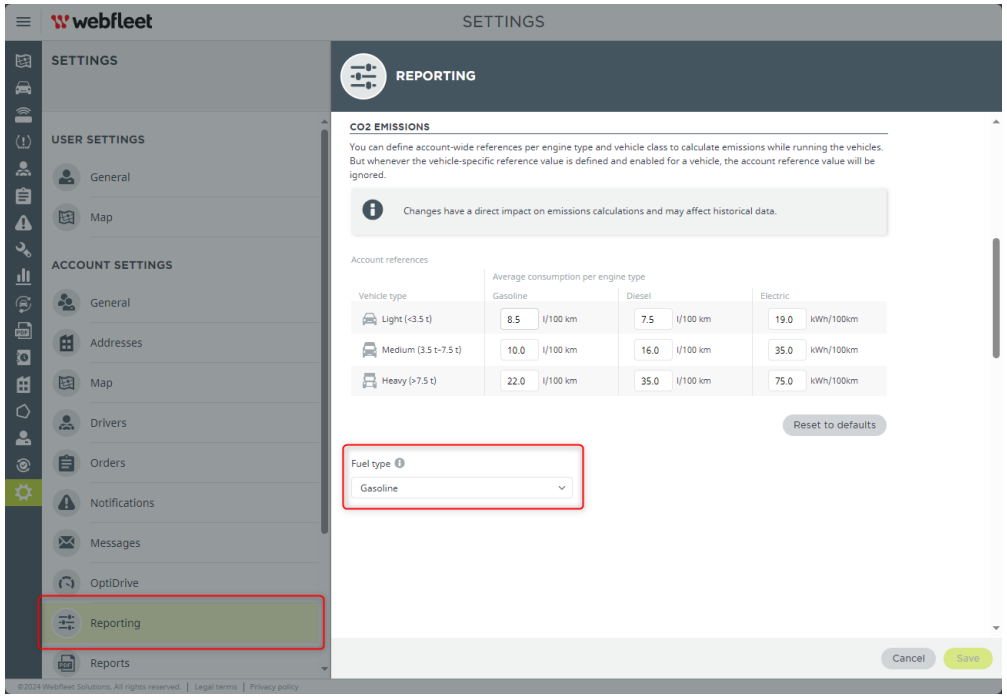
- **Fuel consumption** is reported automatically by the LINK devices or OEM.connect in the vehicle. To properly calculate CO_2 emissions it is necessary to set up a vehicle's fuel type. **Fuel type** can be defined in Webfleet at different levels:

Vehicle level:



If there is no fuel type value defined for a vehicle or account Webfleet will set DIESEL as default for fuel type

Account level: Default fuel type for all vehicles for an entire account (Settings > Reporting > CO_2 emissions)



Edit this field in the Slide Master to add the product name

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Webfleet CO_2 Report: Fuel-based calculation method

Formula :

$$\boldsymbol{CO2\ emission} = fuel\ consumption\ [l] * CO2\ emission\ factor\ [\mathrm{kgCO_2e}/l]$$

For example: If a vehicle consumed **12l** of petrol/gasoline.

The CO_2 emission factor for petrol/gasoline is **2.32 $\mathbf{kgCO_2e}$**

$$CO2\ emission = 12l * 2.32 \frac{kgCO2e}{liter} = 27{,}84\ kg$$

Fuel type	kgCO2e per litre or kilogram of fuel
diesel	2.65
gasoline	2.32
LPG	1.74
CNG	2.79
LNG	2.79

The CO_2 emissions calculations using **real fuel consumption** are irreversible.

Any changes made, such as adjusting the fuel type, affect future data only.

Webfleet CO_2 Report: Distance-based method

The Webfleet CO_2 Report allows customers to create CO_2 emissions reports even if there is no real fuel consumption data available.

It is possible to use *mobile devices or even standalone cameras* as a source of telemetry data. However, reporting CO_2 emissions using distance only requires more input from the user. The following information is required:

- **Fuel type** (as described for the fuel-based method)
- **Average fuel consumption**
- **Vehicle class** (based on the weight of the vehicle)

Webfleet provides default values when information is not provided by the customer. It is crucial to verify and adjust settings to obtain the most accurate results.

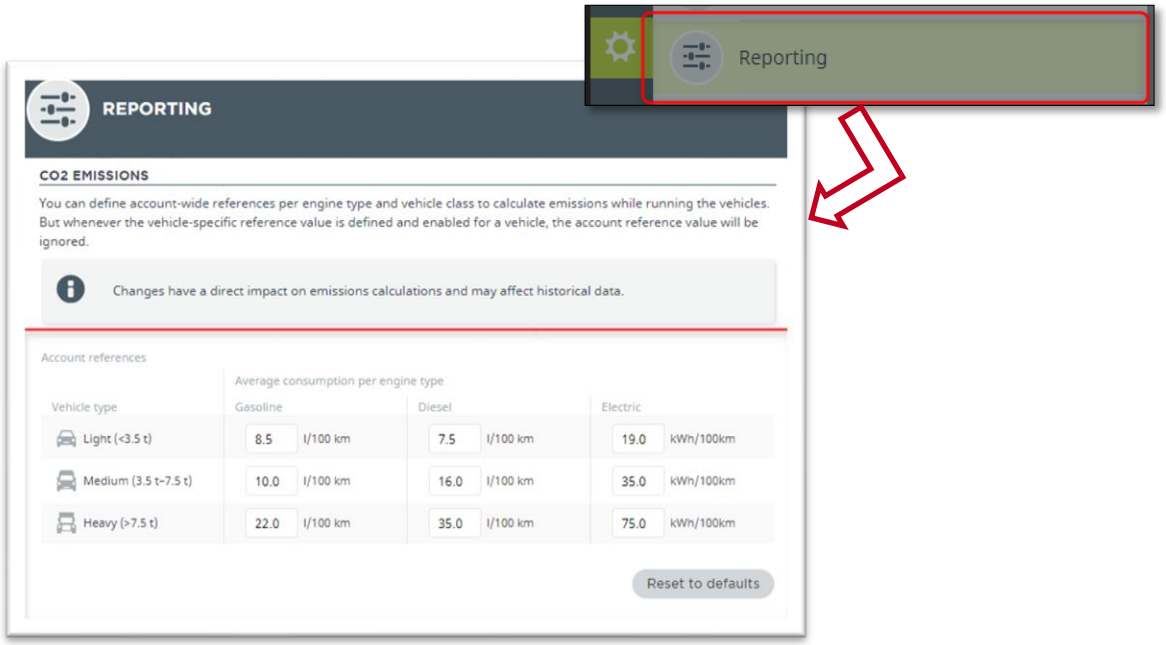
The distance-based method used for CO_2 emissions reporting provides estimated values. It is calculated using the current values of the parameters mentioned above. Any changes related to the fuel type, average fuel consumption or vehicle type will affect the results of the generated CO_2 reports (including historical ones).

The only value that a customer cannot change is the distance (mileage) reported by the vehicle.

Webfleet CO_2 Report: Manual avgerage fuel consumption adjustment

- Average fuel consumption can be defined per vehicle or for the whole account based on vehicle type.
- Account-wide settings support the most common fuel types including petrol, diesel and electric.
- **If a vehicle operates using less common fuel types** (for example LPG, CNG or LNG), then it is required to provide an average fuel consumption value for a vehicle. In Webfleet this can be achieved using the vehicle module and selecting the 'Manual' option when providing fuel consumption:

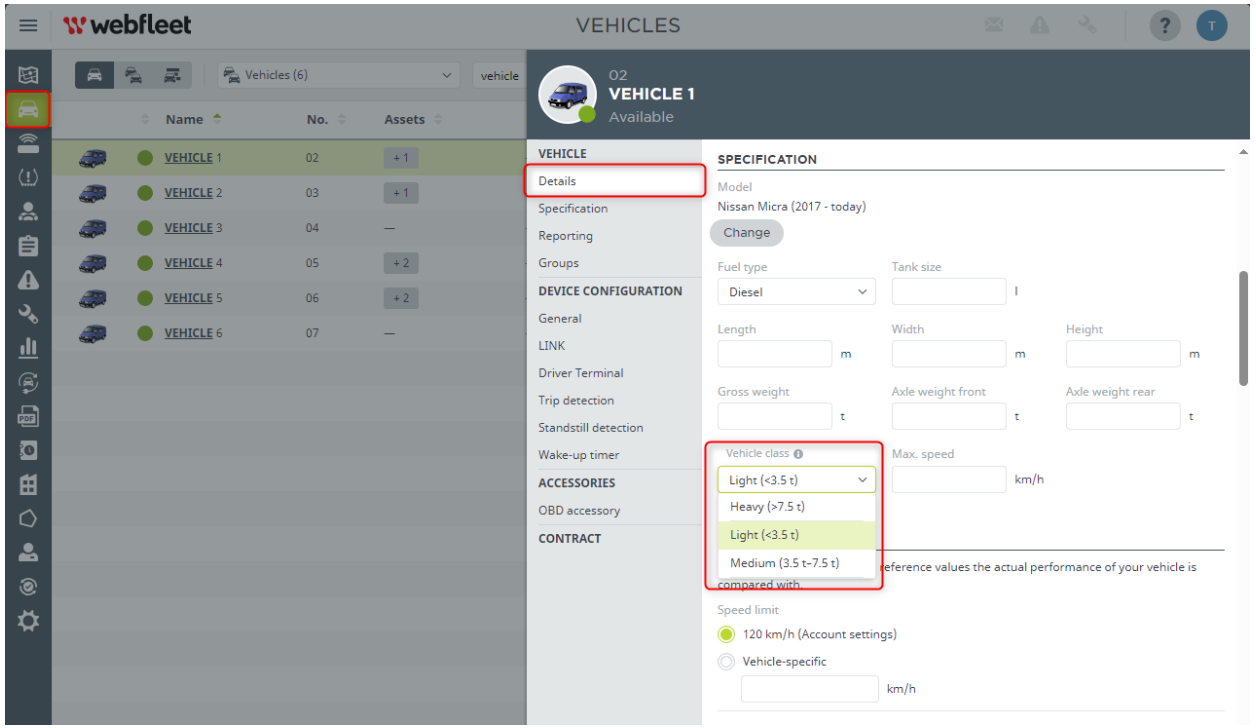
If your fleet uses common fuel types, we can set average fuel consumption on account level for vehicles as shown in the picture below.



Webfleet provides default, values based on global data collected. Setting can be overwritten and adjusted to the fleet characteristics. The 'Reset to defaults' button allows you to revert edited values to default settings.

Webfleet CO_2 Report: Vehicle type selection

Providing the correct vehicle type allows Webfleet to calculate CO_2 emission using the most accurate value of the average fuel consumption. By default, a vehicle is assigned to the class 'Medium' but this can be changed on the vehicle level as shown in the picture below:



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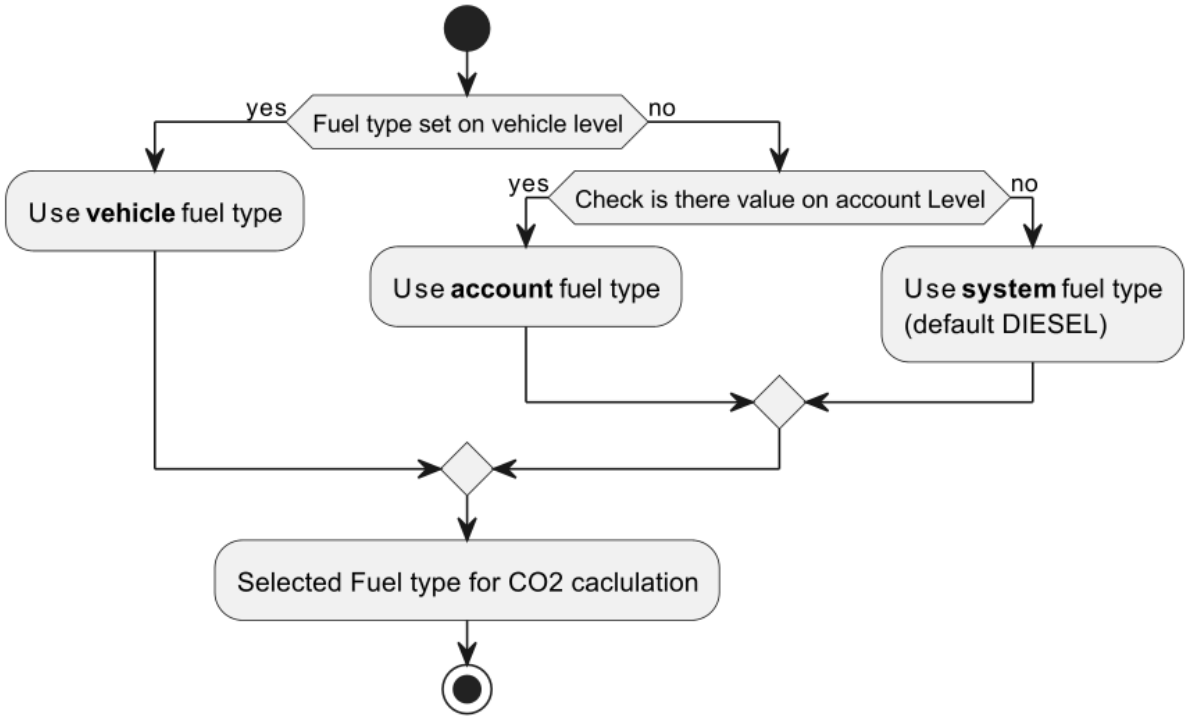
Webfleet CO_2 Report: Fuel type selection

For both methods of CO_2 calculation (fuel-based and distance-based), we use the same method to determine fuel type. Importance of the fuel type data (in order):

- Vehicle level
- Account settings (for the whole fleet)
- System default = DIESEL

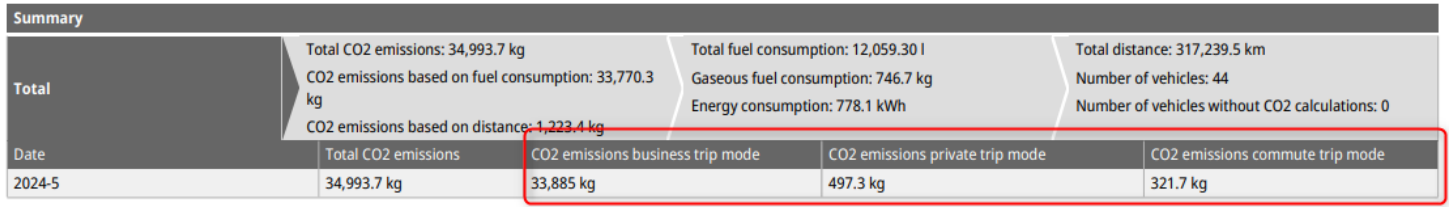
See the diagram for more details.



Trip mode separation and requirements

Similarly like in Webfleet, it is possible to separate different trip modes for CO_2 reports:

- Business
- Commute
- Private



Summary

Total

Total CO2 emissions: 34,993.7 kg
CO2 emissions based on fuel consumption: 33,770.3 kg
CO2 emissions based on distance: 1,223.4 kg

Total fuel consumption: 12,059.30 l
Gaseous fuel consumption: 746.7 kg
Energy consumption: 778.1 kWh

Total distance: 317,239.5 km
Number of vehicles: 44
Number of vehicles without CO2 calculations: 0

Date	Total CO2 emissions	CO2 emissions business trip mode	CO2 emissions private trip mode	CO2 emissions commute trip mode
2024-5	34,993.7 kg	33,885 kg	497.3 kg	321.7 kg

The driver needs to specify the type of trip by using the Webfleet Logbook. The driver can input the type of trip by using these components;

- A Driver Terminal
- PRO 2020 / PRO-i
- Remote LINK Logbook
- Privacy Switch
- Webfleet Logbook
- Webfleet Work App

The fleet manager can change the trip mode after the trip is made using the Logbook module in Webfleet.

Webfleet CO_2 emission for EVs

EVs are also part of the CSRD reporting standard, depending on the method of electricity generation.

Similarly to combustion engines, reports can be created using energy consumption-based or distance-based methods**.** The energy powering electric vehicles primarily comes from the electricity grid, which is increasingly being supplemented by renewable energy sources. Grid electricity is generated by various types of power plants.

Electric Vehicles can be charged using numerous different methods and from several sources. Webfleet cannot identify the actual method of electricity generation or the quality of electricity.

From reports, scopes are separated:

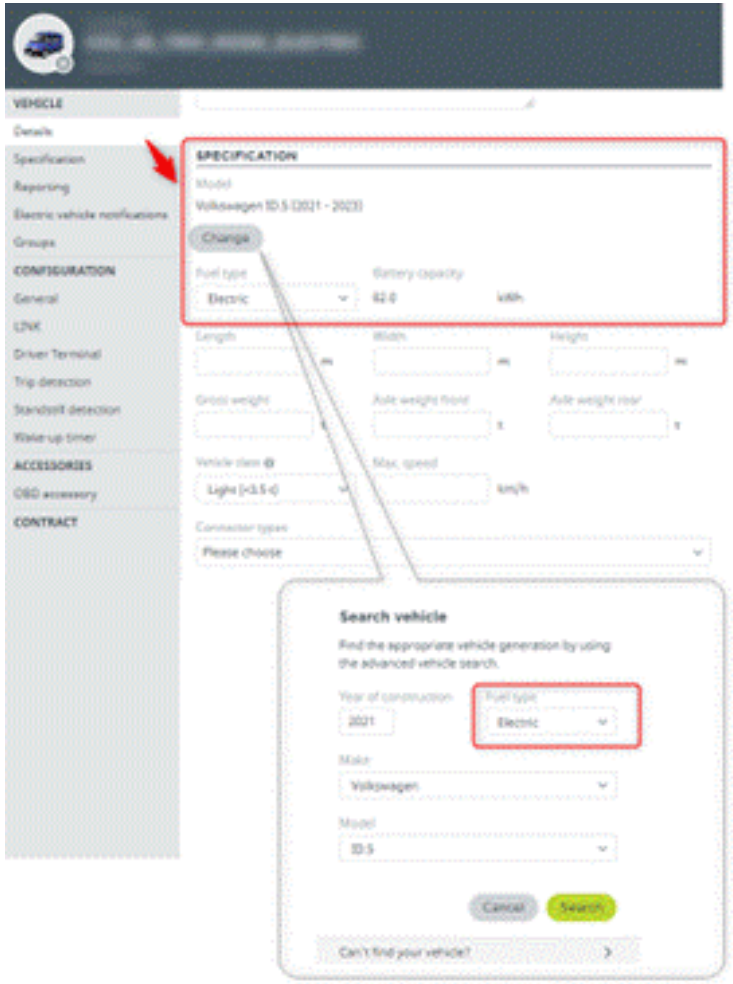
- Scope 1 direct = emission from vehicle tailpipe
- Scope 2 indirect = purchased energy/electricity grid

Due to the above reasons, we have developed two possibilities for making EV CO_2 reporting possible, which are further explained in this report; those methods are:

- To use official generalized country grid electricity factors.
- Allow users to select their own factors based on better personal knowledge

National grid values

- Since the CO_2 emission factor for electricity production varies by country, using country-specific grid values is essential.
- To calculate energy consumption and CO_2 emissions for EVs , the vehicle's engine type must be set to electric. This can be verified by setting the proper vehicle make/model.
- Current factors are from the EEA (European Environment Agency) to provide country-specific Greenhouse gas emission intensity of electricity generation values.
- **NOTE:** The source of the data may change in the future.



CO_2 Emission calculating formula for EV

Electricity Consumption: Determine the EV's energy consumption, which is usually measured in kilowatt-hours (kWh) per 100 kilometres (kWh/100 km).

Formula:

$$Total\ energy\ consumption\ [kWh] = Energy\ for\ driving + Energy\ for\ others - Energy\ recovered$$

- Energy for others = energy used to power vehicle systems like, for example, air conditioning (other than driving)
- Energy recovered = energy the vehicle recovered (for example, from breaking / recuperation)

Electricity Emission Factor: Obtain the CO_2 emission factor for the electricity consumed, measured in grams of CO_2e per kWh (g CO_2e/kWh). This factor varies by region and the energy sources used in the grid (e.g., coal, natural gas, nuclear).

$$CO2\ Emissions\ (g/km) = Electricity\ Consumption\ (kWh/km) \times Electricity\ Emission\ Factor\ (g\ CO2e/kWh)CO2$$

CO_2 Emission for EVs

To calculate CO_2 emissions for a specific Electric Vehicle, we need to know the CO_2 emission factor for each kWh of energy used for charging the vehicle.

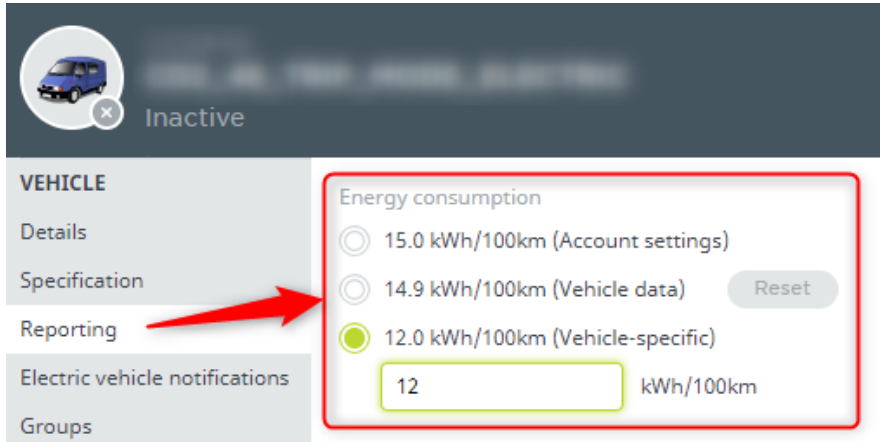
Logic:

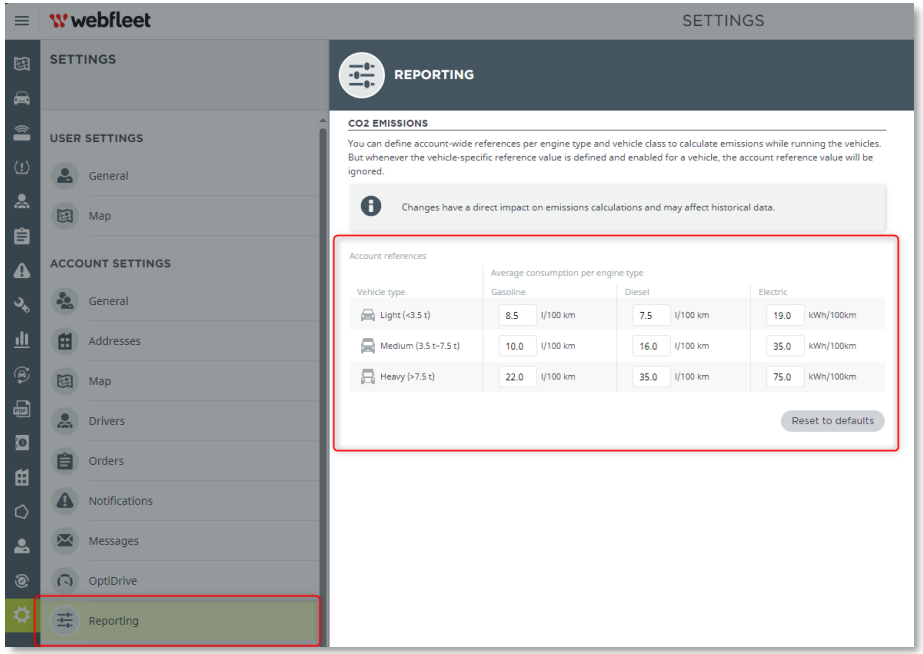
- We need to assume energy used for driving that we know its basis for CO_2 emissions calculation (we are aware the charging efficiency is around 90% using AC chargers / some parasitic loss)
- Only about 50% of EVs are offering actual energy consumption. We will use the account's customer country to select the value of the CO_2 per kWh energy produced factor (for the distance-based method)
- If the vehicle does not report energy consumption we will use the <u>average</u> kWh <u>per vehicle class</u> - values based on Webfleet (light, medium) and truck producer's data for heavy fleets.
- To calculate CO_2 emission per trip, we will use **the start position country** and CO_2 emission factor per kWh produced energy on the country level.
- A user can set its own CO_2 emissions factor per kWh energy produced (including value 0 - for example, for solar power or countries outside the EU that do not calculate CO_2 for EVs)

Average energy consumption

Similarly, like with fossil fuels, energy consumption for EVs can be defined for individual objects or whole accounts based on vehicle class.

In Webfleet this can be achieved using the vehicle module and selecting the “Manual” option when providing energy consumption:





Users can set average energy consumption on account level for vehicles.

Webfleet provides default values based on solid data experience and extensive global data sets.

The setting can be overwritten and adjusted to fit the fleet's characteristics. The *“Reset to defaults”* button allows you to revert edited values to default settings.

Energy-based calculation example

The emission factor to produce 1kWh of energy for Germany is 366g CO_2 (this value is changing over time because of global energy market changes)

$$CO2\ emission = \ energy\ used\ [kWh] * CO2\ emission\ factor\ \left[\frac{\mathrm{gCO_2e}}{kWh}\right]$$

Example Germany: a vehicle consumes **23kWh** of electric energy during the trip. The CO_2 emission to produce 1kWh of energy is **366 gCO_2e**

$$CO2\ emission = \ 23kWh * 366\ \frac{gCO2e}{kWh} = 8{,}418\ kg$$

CO_2 emissions calculated using real energy consumption are immutable. Any changes, such as adjusting the CO_2 emission factor per kWh of energy produced, affect future data only.

Distance Based calculation for EV

If energy consumption is not available, then the **distance-based** method is used.

Get average energy consumption value

- Defined on vehicle level or,
- Account level per vehicle class or,
- System default value per vehicle class.

Example of calculating energy consumption using distance and average energy consumption:

$$energy\ consumption\ [kWh] = \frac{avg.energy\ consumption\ [kWh\ per\ 100km]}{100} * distance\ driven\ [km]$$

For example, A vehicle travelled 250 km. The average energy consumption is 19 kWh/100km.

$$energy\ consumption\ [kWh] = \frac{19kWh}{100\ km} * 250\ km = 47{,}5\ kWh$$

The distance-based method for CO_2 emissions reporting provides estimated values.

It is calculated using the current values of the parameters mentioned above. Any changes related to average energy consumption, vehicle class, CO_2 emission factor etc. will affect the results of the generated CO_2 reports (including historical ones).

The only immutable value is the distance reported by the telematic device.

Customer benefits

- **Real-time and historical insights**
  - The Webfleet CO_2 Report enables you to accurately measure the CO_2 produced by your fuel or electric vehicles. You can see not only how much CO_2 your vehicles are emitting, but also why and where it is happening. With these real-time and historical insights, you can zero in on where the real opportunities are to cut your fleet's emissions.

- **The impact of driving behaviour**
  - Driving behavior like idling and harsh braking can directly lead to needless fuel consumption and CO_2 emissions. With data on CO_2 emissions in your Webfleet OptiDrive 360 reporting, you can see specifically where drivers are needlessly increasing your carbon footprint and coach them to adopt more sustainable and cost-effective driving styles.

- **Report accurately and easily on your business' CO_2 emissions and environmental impact**
  - With the Webfleet CO_2 Report you can rest assured you'll have all the data you need to report on your CO_2 emissions – both to the authorities and to your customers that request it. It's the only CO_2 reporting tool from a telematics provider with a methodology certified by TÜV Rheinland[1]. So, whether you need this data to stay compliant, stay competitive or both, Webfleet has you covered.

Tariff requirements

- Webfleet CO_2 reports are available to all Webfleet customers with an active Webfleet subscription.
- The Webfleet CO_2 Report fuel-based calculation method is available to customers on ECO, NAV, and LIVE subscriptions.
- The Webfleet CO_2 Report distance-based calculation method is available to customers who do not get fuel data due to having a TRACK or LINK tariff or because their device does not report fuel.
- The best calculation method for CO_2 emissions values is the fuel-based method. If the customer would like to get their real fuel consumption values, they will need to have an ECO or above tariff. It will provide actual fuel consumption data, allowing them to:
    - Understand their current consumption level
    - Set reduction targets
    - Track how they are performing versus targets.
- Both calculation methods are available in the Webfleet CO_2 Report (monthly and annual)
- The Webfleet CO_2 Report is available for combustion engines and electric vehicles.
- Electric vehicles must also be reported based on their Scope 2 emissions. Scope 2 emissions are indirect emissions from the energy/electricity grid purchased.
- Private and business trip input can be recorded by the driver using these components: A Driver Terminal, PRO 2020 / PRO-I, Remote LINK Logbook, Privacy Switch, Webfleet Logbook, and Webfleet Work App.
- Fleet managers can change the **trip mode** after the trip is made using the Logbook module in Webfleet.

FAQs:

What does the TÜV Certification mean?

- To meet state-of-the-art scientific standards, all platform-embedded calculations and decarbonisation solutions are fully aligned with internationally recognised scientific methodologies and standards such as the Greenhouse Gas Protocol and the Science Based Targets initiative (SBTi). The scientific accurateness of the applied Corporate Carbon Footprint (CCF) calculation methodology, Scope 1, is certified by TÜV Rheinland.

What is the Corporate Sustainability Reporting Directive (CSRD)?

- The Corporate Sustainability Reporting Directive (CSRD) is a new EU regulation that requires companies to report on risks and opportunities arising from social and environmental issues, such as climate change, and on the impact of their activities on people and the environment. While different types of businesses will be affected by the regulation at different times, over the coming years many thousands of companies across the continent will need to comply with it.

Why is CO_2 tracking important?

- Accurately tracking CO_2 emissions is crucially important for any fleet of any size for many reasons. Most importantly, if you do not have visibility over the level of CO_2 you produce you cannot take meaningful action to reduce your emissions or track the success of such actions over time. So, any attempt to run more sustainably or efficiently relies on this visibility. As mentioned above, there is also the need to comply with regulations. More and more companies will have to provide accurate data on CO_2 emissions over the coming years or risk penalties and potential loss of earnings. So, for the planet, for business and for your company's bottom line, tracking CO_2 is no longer an optional activity but a must-do.

FAQs:

How can I use the Webfleet CO_2 Report for my fleet?

- The Webfleet CO_2 Report gives you accurate information on the level of CO_2 produced across your fleet. With this data you can fulfil your reporting responsibilities on things like CSRD, identify areas in your operation where you could take action to run more sustainably and measure the success of your initiatives to reduce your impact on the planet.

How can I reduce my carbon footprint using fleet management software?

- Connecting your fleet with a fleet management solution like Webfleet is the first step to getting the kind of insights into your CO_2 emissions you need to start reducing your impact on the environment. Improving driving behaviour, making the switch to EVs, reporting on CO_2, detecting tyre underinflation, optimising your routes, keeping your vehicles as fuel efficient as possible – all of this and much more becomes possible with the power of Webfleet at your fingertips.

What is Scope 3 GHG emissions?

- Scope 3 GHG emissions are all indirect emissions that occur in the upstream and downstream value chain of the reporting company. Scope 3 emissions are typically the greatest component of a corporate or financial services company's footprint. They are also difficult to measure and manage because they occur outside of an organisation's operational control.

FAQs:

How could smaller organizations be affected by CSRD and Scope 3 emissions?

- Even if your business is not obliged to report on your own carbon footprint, you might be asked by your larger customers to provide information on the CO_2 emission from your vehicles when providing goods or services to them. This is because they will need to report on those Scope 3 emissions to comply with CSRD. This may mean the ability to provide accurate data on your CO_2 can become a competitive advantage for you in the coming years.

How can Webfleet help you to provide information on your carbon footprint?

- The Webfleet CO_2 Report gives you accurate information on the CO_2 produced by your connected vehicles. With this data, you can fulfil your CSRD reporting responsibilities or provide this data to your larger customers, so they can report on their Scope 3 emission.

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